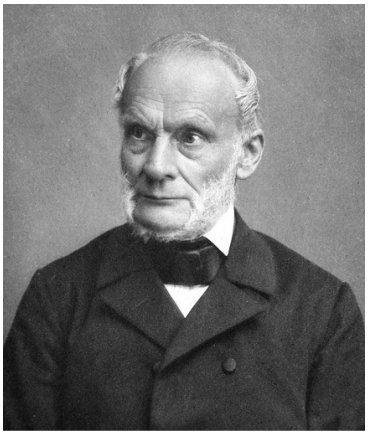
Perhaps poetry is another of science's deepest roots: the capacity to see beyond the visible. Carnot names his first son after Saadi. Sadi Carnot is thus born out of poetry and rebellion.

As a young man, he develops a passion for those steam engines that at the start of the nineteenth century are beginning to transform the world by using fire to make things turn. In 1824, he writes a pamphlet with the alluring title "Reflections on the Motive Power of Fire," in which he seeks to understand the theoretical basis of the functioning of these machines. The little treatise is packed with mistaken assumptions: he imagines that heat is a concrete entity —a kind of fluid that produces energy by "falling" from hot things to cold, just as the water in a waterfall produces energy by falling from above to below. But it contains a key idea: that steam engines function, in the final analysis, because the heat passes from hot to cold.



Sadi's pamphlet finds its way into the hands of a fierce-eyed, austere Prussian professor called Rudolf Clausius. It is he who grasps the fundamental issue at stake, formulating a law that was destined to become famous: if nothing else around it changes, heat *cannot* pass from a cold body to a hot one.

The crucial point here is the difference from what happens with falling bodies: a ball may fall, but it can also come back up, by rebounding, for instance. Heat cannot.

This is the *only* basic law of physics that distinguishes the past from the future.

None of the others do so. Not Newton's laws governing the mechanics of the world; not the equations for electricity and magnetism formulated by Maxwell. Not Einstein's on relativistic gravity, nor those of quantum mechanics devised by Heisenberg, Schrödinger, and Dirac. Not those for elementary particles formulated by twentieth-century physicists. . . . *Not one* of these equations distinguishes the past from the future.[11] If a sequence of events is allowed by these equations, so is the same sequence run backward in time.[12] In the elementary equations of the world,[13] the arrow of time appears *only* where there is heat.[*] The link between time and heat is therefore fundamental: every time a difference is manifested between the past and the future, heat is involved. In every sequence of events that becomes absurd if projected backward, there is something that is heating up.

If I watch a film that shows a ball rolling, I cannot tell if the film is being projected correctly or in reverse. But if the ball stops, I know that it is being run properly; run backward, it would show an implausible event: a ball